

Signature Algorithms		Handshake Latency [ms] — Lower is Better																																																	
		5										10										100										500																			
MLDSA44	FALCONPADDED512	140.8	132.3	79.2	72.9	71.5	29.1	26.0	26.3	21.3	18.8	20.1	16.2	17.3	14.9	11.6	11.8	9.9	10.3	10.1	8.5	6.5	8.5	7.1	8.2	7.6	5.8	7.3	6.9	4.7	6.1	5.7	5.4	5.4	4.8	4.3	3.9	3.9	4.0	4.2	4.0	3.2	2.5	2.4	2.2	2.6	2.9				
	FALCON512	141.3	132.5	79.9	72.9	71.4	29.3	26.2	26.8	21.1	19.6	18.4	16.3	16.3	14.5	11.7	11.7	10.6	10.1	10.3	8.8	6.4	8.7	7.7	8.5	7.7	6.0	7.0	6.8	4.8	6.2	5.7	5.4	5.5	4.9	4.5	3.6	4.0	4.0	3.8	3.7	3.0	2.6	2.7	2.4	2.6	2.5				
	FALCONPADDED1024	139.9	134.5	78.2	73.6	71.4	29.1	26.2	26.4	21.4	21.5	17.6	16.3	15.8	14.3	11.8	11.7	10.2	10.9	10.3	8.9	6.7	9.0	7.2	8.5	7.8	5.7	7.1	6.8	4.7	6.1	5.7	5.5	5.5	4.8	4.5	3.8	3.9	4.1	3.8	3.7	3.0	2.9	2.6	2.8	2.7	2.6				
	MLDSA65	141.3	131.8	78.9	74.0	71.6	29.3	26.6	26.8	21.2	19.5	19.0	16.9	15.5	15.2	11.8	12.1	11.2	9.8	11.1	8.8	6.8	8.8	7.9	8.7	7.8	5.8	7.5	7.5	4.8	6.7	6.0	5.7	5.9	5.0	5.3	4.1	4.0	4.2	4.4	4.3	3.6	3.0	2.9	2.8	3.4	3.0				
	FALCON1024	141.1	134.0	78.4	73.3	71.6	29.4	26.4	26.8	21.2	21.7	19.1	17.0	15.8	14.3	12.0	11.9	10.1	10.9	10.7	8.8	6.9	9.0	7.3	8.7	8.0	6.2	7.4	7.1	5.6	6.3	5.9	5.6	5.7	5.1	4.8	4.1	4.3	4.2	4.1	3.8	3.7	3.3	3.1	3.0	3.0					
	P256 + FALCONPADDED512	141.2	133.4	78.9	73.4	72.5	29.6	27.0	26.7	22.1	19.1	18.4	16.8	15.7	15.1	11.2	12.0	10.3	10.7	10.6	9.0	7.4	9.0	7.7	8.7	8.4	6.1	7.3	7.3	5.6	6.5	6.1	6.2	5.9	5.8	5.2	4.5	4.6	4.3	4.3	4.2	3.6	3.3	3.5	3.4	3.3	3.3				
	P256 + MLDSA44	140.2	134.7	79.0	73.6	72.1	29.5	26.8	26.5	21.0	21.6	20.3	16.6	16.7	13.8	11.5	11.9	10.5	10.8	10.6	9.0	7.0	8.8	7.6	8.8	8.3	6.2	7.2	7.3	5.4	6.5	6.0	6.0	5.9	5.6	5.2	4.6	4.5	4.3	4.4	4.2	3.6	3.6	3.4	3.2	3.4	3.3				
	P256 + FALCON512	142.1	134.7	78.5	73.4	72.1	29.5	26.8	26.7	21.4	21.9	18.9	16.6	15.2	14.5	11.3	12.0	10.3	10.9	10.8	9.3	7.1	9.0	7.7	9.0	8.4	6.3	7.2	7.1	5.5	6.5	6.1	6.3	6.0	5.6	5.2	4.4	4.7	4.5	4.2	4.2	3.8	3.3	3.1	3.4	3.3	3.5				
	MLDSA87	144.1	133.0	79.1	75.5	71.9	29.9	26.9	27.3	21.1	21.2	18.2	17.1	16.2	15.0	11.6	12.3	10.7	11.0	11.1	9.8	7.0	9.0	7.7	8.9	8.3	6.4	7.5	7.6	5.4	7.1	6.1	6.1	6.4	5.4	5.5	4.7	4.4	4.4	4.4	4.7	3.5	3.3	3.1	3.2	3.2	3.4	3.9			
	P384 + MLDSA65	142.8	135.9	81.6	74.2	73.2	30.6	27.9	28.3	22.6	21.2	20.2	17.8	17.0	14.9	13.2	13.3	11.6	12.5	12.0	10.1	9.0	9.9	8.7	9.8	9.4	7.8	8.7	8.4	6.8	7.6	7.2	7.2	6.9	6.9	6.4	5.5	5.7	5.6	5.5	6.0	5.1	4.7	4.8	4.8	5.0	4.9	4.9			
RSA3072 + FALCONPADDED512	RSA3072 + FALCONPADDED512	142.3	134.3	80.4	74.6	73.5	30.9	28.1	28.1	23.7	20.2	23.1	18.2	18.0	17.3	14.9	13.4	12.2	14.4	12.3	10.7	9.9	10.5	9.7	10.2	9.6	8.5	8.9	8.8	7.1	8.1	7.5	7.4	7.3	6.9	6.7	5.6	6.0	5.7	5.5	5.4	5.5	4.8	4.7	4.7	4.8	5.5				
	RSA3072 + MLDSA44	143.9	135.0	80.5	74.9	73.4	30.9	28.1	27.9	22.2	21.2	21.1	18.1	18.7	17.3	14.6	13.6	12.0	13.3	11.9	10.8	10.6	10.4	9.0	10.0	9.6	8.5	8.9	8.7	7.4	7.7	7.3	7.4	7.3	7.0	6.5	5.6	5.8	5.5	5.4	5.3	5.3	5.3	5.2	5.2	5.6					
	RSA3072 + FALCON512	143.2	136.6	80.5	75.3	73.6	31.0	28.1	27.9	23.0	20.1	20.5	18.0	17.8	17.3	15.6	13.5	11.9	13.9	12.2	11.5	9.7	10.5	8.9	10.1	9.6	9.3	8.8	8.6	7.8	7.8	7.4	7.7	7.5	6.9	6.5	5.8	5.8	5.7	5.4	5.4	5.7	4.8	4.7	4.9	5.0					
	P521 + MLDSA87	143.2	136.9	81.8	76.3	75.2	32.1	29.4	29.3	24.2	22.9	22.4	19.2	19.4	18.7	15.9	14.9	12.8	14.4	13.3	12.1	11.4	11.2	10.6	10.1	9.9	9.7	9.2	9.0	8.8	8.6	8.3	7.9	7.3	7.3	7.2	6.9	6.8	6.2	6.1	6.2	5.9	6.4	6.0	6.0						
	P521 + FALCONPADDED1024	143.8	135.2	82.0	75.9	75.1	32.3	29.6	29.5	24.2	20.6	20.3	19.8	19.2	19.8	16.7	15.0	13.0	16.2	13.4	11.9	12.4	11.7	11.0	11.4	11.3	10.1	10.2	9.9	9.4	9.3	8.8	9.0	8.8	8.5	8.2	7.1	7.5	7.3	7.0	7.1	6.3	6.4	6.2	6.3	6.1					
	P521 + FALCON1024	150.0	136.3	82.3	76.2	75.0	32.3	29.6	29.5	23.9	23.3	23.8	19.8	18.0	19.1	16.9	14.7	14.0	15.3	13.4	12.2	11.6	11.7	10.4	11.6	11.2	10.5	10.3	10.1	10.0	9.5	8.9	9.0	8.7	8.5	8.1	7.2	7.5	7.2	6.9	7.0	6.3	6.3	6.2	6.4	6.2					
	SPHINCSSHA2128FSIMPLE	154.7	137.6	87.4	79.1	77.5	35.7	32.1	32.2	26.6	28.5	28.1	22.5	23.9	26.1	22.3	17.9	15.8	21.5	16.7	15.0	18.2	14.7	14.2	14.7	13.6	17.3	13.3	13.0	13.9	12.2	11.8	11.4	11.7	11.1	10.7	9.8	10.0	10.4	9.9	9.8	9.2	8.9	9.1	9.0	8.7	8.9				
	P256 + SPHINCSSHA2128FSIMPLE	157.7	137.9	93.4	80.0	78.3	35.7	32.8	32.8	31.0	28.9	26.9	22.8	28.2	26.3	23.6	18.1	17.3	22.9	16.8	15.1	19.0	15.1	13.7	14.9	14.3	16.1	13.4	13.3	15.9	12.3	12.1	12.4	12.3	11.8	11.3	10.7	10.6	10.6	10.2	9.5	9.8	9.8	9.5	9.7	9.5					
	RSA3072 + SPHINCSSHA2128FSIMPLE	148.0	139.5	87.1	82.1	79.8	37.2	34.2	34.3	30.1	30.8	34.4	24.1	31.4	24.9	26.1	19.7	19.8	22.8	18.1	16.9	19.6	16.5	15.1	16.6	15.9	20.7	14.8	14.7	17.7	13.9	13.5	13.6	13.4	13.2	12.8	12.1	11.8	12.1	12.0	11.5	11.2	11.0	10.9	10.7	10.9	11.1				
	SPHINCSSHA2192FSIMPLE	151.5	143.6	91.4	84.9	81.9	39.8	36.7	36.9	32.5	41.0	38.8	27.3	35.9	33.7	31.2	22.2	21.7	26.0	20.9	19.5	25.7	19.2	17.9	19.2	18.3	23.0	17.6	17.3	23.9	16.5	16.8	15.9	16.0	15.6	15.3	14.1	14.2	14.8	15.0	14.1	13.4	13.5	13.4	13.5	13.3	13.6				
P384 + SPHINCSSHA2192FSIMPLE	152.2	143.7	91.2	86.7	83.8	41.2	38.5	38.2	33.4	32.3	28.7	28.5	37.4	31.5	28.6	23.6	25.0	26.6	22.4	20.6	28.8	20.4	19.1	20.3	20.7	27.3	19.2	18.8	23.4	17.9	18.1	18.9	18.0	17.4	17.0	16.8	16.2	16.3	15.9	15.7	15.4	15.1	16.3	15.1	15.4	14.9					
SPHINCSSHAKE128FSIMPLE	157.6	161.4	104.4	87.0	86.4	42.4	39.4	39.2	35.3	27.4	27.3	29.4	25.7	35.1	31.1	24.8	23.7	34.0	23.5	22.0	26.8	21.4	21.7	21.7	20.9	29.8	19.9	19.8	29.0	19.2	19.0	18.6	18.7	18.9	18.1	17.0	17.1	17.3	17.1	16.7	16.5	16.0	16.0	15.8	15.8	15.9					
P256 + SPHINCSSHAKE128FSIMPLE	156.6	159.8	93.9	87.5	87.5	42.4	40.0	39.6	35.4	27.8	28.0	29.6	26.3	34.1	32.7	25.1	24.3	35.4	23.7	22.0	31.6	21.8	21.6	21.7	21.4	30.1	20.1	20.2	29.5	19.4	19.1	19.1	19.0	19.0	19.5	18.0	17.8	17.8	17.8	17.5	17.1	17.0	17.8	17.0	16.9	16.9					
RSA3072 + SPHINCSSHAKE128FSIMPLE	156.3	163.6	95.5	88.9	89.3	44.1	41.2	40.9	36.3	30.2	29.2	31.2	27.7	25.8	38.9	26.3	25.7	29.3	25.0	23.8	30.6	23.4	22.3	23.2	23.0	31.9	21.9	21.5	31.7	20.8	20.6	21.1	20.9	19.9	20.5	19.2	18.8	19.0	18.5	19.2	18.0	18.4	18.4	18.3	18.3						
SPHINCSSHA2256FSIMPLE	160.1	169.3	104.2	95.2	92.2	50.0	46.9	47.4	42.6	36.2	35.1	37.8	42.4	31.3	30.2	32.1	31.8	30.0	31.6	29.9	38.4	29.8	28.0	28.9	28.3	30.0	27.8	27.6	34.1	27.0	26.7	26.1	26.1	25.5	26.5	24.2	24.7	24.9	24.6	24.5	24.7	23.7	23.7	23.4	23.8	23.3					
SPHINCSSHAKE192FSIMPLE	162.9	173.2	101.1	95.6	95.2	50.9	47.6	47.5	42.4	37.2	35.6	37.8	32.8	31.5	31.9	33.1	33.4	29.7	32.2	30.3	27.7	30.1	29.7	29.5	28.9	31.9	28.6	28.5	25.9	27.2	28.1	26.3	27.0	26.2	27.0	25.9	24.9	26.1	25.4	25.3	24.2	24.0	24.2	23.8	24.4	24.1					
P384 + SPHINCSSHAKE192FSIMPLE	161.4	171.4	117.0	97.4	97.9	52.8	49.4	48.9	44.7	37.5	37.0	39.4	36.1	34.5	34.1	34.5	34.8	31.2	33.7	31.6	30.1	31.0	30.4	31.0	30.9	28.2	30.1	29.6	27.8	28.4	29.9	30.1	28.3	27.9	29.0	28.1	26.9	27.2	27.3	26.7	26.1	26.0	26.7	26.9	25.6	26.9					
P521 + SPHINCSSHA2256FSIMPLE	162.9	183.7	103.5	98.0	95.6	53.7	50.5	50.4	54.2	40.5	39.2	37.8	40.0	37.9	41.7	33.9	35.2	36.6	33.0	34.5	32.7	30.5	32.8	31.2	31.7	32.4	29.3	30.8	30.3	29.3	29.7	29.4	29.5	29.5	29.2	29.9	28.6	28.4	27.8	27.5	27.3	27.0	27.3	26.7	27.8	26.5					
SPHINCSSHAKE256FSIMPLE	200.4	203.4	124.4	119.8	118.9	73.5	70.0	69.4	68.4	58.5	58.1	61.1	57.1	54.4	53.4	54.7	54.8	51.6	54.2	52.3	50.0	52.4	51.0	51.9	51.5	49.4	50.2	50.3	48.5	49.5	50.4	49.2	50.0	48.2	48.1	47.5	47.0	48.1	47.2	47.4	46.5	47.3	45.9	47.1	46.4	46.3					
P521 + SPHINCSSHAKE256FSIMPLE	197.7	199.4	130.7	123.6	121																																														